Xi Li

Machine Learning Researcher — Multimodal ML and Research Engineering

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Seeking Research Internships

Profile

Ph.D. researcher developing multimodal models, large-scale data pipelines, and reproducible evaluation systems. Experience spans image-to-molecular prediction, interpretable LLM reasoning, and generative-AI products deployed in real production workflows.

Education

University of California, Irvine	*Ph.D., Computer Science*	Sep 2023 – Present
Tsinghua University	*M.S., Computer Science and Technology*	Sep 2019 – Jun 2022
University of Science and Technology of China	*B.S., Information Security*	Sep 2015 – Jun 2019

Expected Ph.D. graduation: December 2027.

Research Experience

University of California, Irvine — **Irvine, CA**

Ph.D. Student Researcher — *Sep 2023 – Present*

- **STMoE — multimodal prediction pipeline.** Led development of a dual-scale model for **single-cell gene-expression prediction across 25 organs**, combining frozen visual encoders, soft contrastive alignment, and organ-routed retrieval. Kept reference expression profiles strictly within the training partition.
- **Cross-sample generalization.** Retained **70% of predictive performance** when generalizing to unseen samples.
- **Data engineering and leakage control.** Built cell-aligned data and benchmarks for **CellST-24M** and **sMMC-22M**; tracked study/slide provenance, assay, and coordinate metadata, distinguishing native Xenium cells from derived Visium HD targets. The **expanded sMMC collection** contains **28.3M cell-aligned targets across 25 tissue categories**; the formal paper retains the frozen-release name sMMC-22M.
- **TargetSage — interpretable drug-target discovery (co-author).** Responsible for **disease-state verification and multi-target investigation** within an evidence-grounded LLM-reasoning framework. The overall model improved **macro-average Adjusted F1 by 37% relative to the strongest of 10 baselines**, across 15 tasks and 19,032 genes.
- **Extension to spatial perturbation modeling.** Extended the TargetSage reasoning approach to spatial perturbation research, connecting disease-state context and multi-target hypotheses with perturbation-conditioned tissue responses.
- **Spine — spatial response modeling (first author).** Developed a formulation that separates tissue context from perturbation-conditioned response fields in spatial genomics; manuscript under review.

Multimedia Lab, Tsinghua University — **Beijing, China**

Researcher, AI Broadcast Production — *Oct 2021 – Sep 2022*

- Built an audio–visual cross-attention system for automated camera switching, crop selection, and virtual camera motion; **deployed at China's National Centre for the Performing Arts** for live orchestral production.

Technical Skills

Modeling Multimodal learning, transformers, mixture-of-experts, contrastive learning, LLM reasoning, positive–unlabeled models

Engineering Python, PyTorch, multi-GPU experimentation, image–omics alignment, dataset QC, grouped evaluation and reproducible benchmarking

Industry and Product Experience

Hantao Zhiying **China**
Chief Product Lead *2022 – 2023*
- Owned AI imaging product definition, user experience, and delivery coordination across research, engineering, and design.

ByteDance AI Lab **Beijing, China**
AI Product Manager Intern *May 2021 – Dec 2021*
- Incubated **Alchemy**, an internal image-generation platform; drove product scoping and internal launch for **Fanqie Novel book-cover creation**.
- Supported the **Chimera** digital-human project, owning automotive-client engagement and solution sales for **digital-human in-car assistants**.

ByteDance — Feishu/Lark **Beijing, China**
Data Product Intern *Jun 2020 – Nov 2020*
- Used behavior analytics and relevance analysis to inform enterprise document and knowledge-search product decisions.

Nolibox / Computational Aesthetics **Beijing, China**
Partner and Co-founder *Mar 2020 – Aug 2020*
- Developed automatic logo-generation and font-stroke generation tools for AI-assisted design.

Company milestone (post-tenure): Series A financing of tens of millions of RMB (2024).

Mentoring and Open-Source Leadership

Google Summer of Code (GSoC) **2025 & 2026**
Project Mentor — UC Santa Cruz Open Source Program Office
- Provided **end-to-end mentorship to two students** across the 2025 and 2026 projects, guiding problem formulation, implementation, experiments, and paper development; both projects have code repositories.
- Guided a student to publish **one student-first-author paper at BIBM**.
- Projects: **2026: Omni-ST** — instruction-driven multimodal spatial modeling; **2025: Disentangled Generation and Editing of Pathology Images** — controllable pathology-image generation and editing.

Selected Publications and Manuscripts

Publication links: xenali96.github.io/publications.html.

Published / accepted

1. **Xi Li**, Yaqi Hu, Ziheng Duan, Chaoyang Wang, and Jing Zhang. "STMoE: Multi-Scale Mixture-of-Experts for Single-Cell Gene Expression Prediction from Histology." **CIKM 2026**, accepted. *First author.*
2. **Xi Li**, Yaqi Hu, Ziheng Duan, Diptanshu Sikdar, Yang Liu, and Jing Zhang. "sMMC-22M: A Context-Aware Dataset and Benchmark for Single-Cell Spatial Transcriptomics." **NeurIPS 2026**, accepted poster. *First author.*
3. Ziheng Duan*, **Xi Li***, Zhiqing Xiao, Rex Ying, and Jing Zhang. "MUSE: A Multi-Slice Joint Analysis Method for Spatial Transcriptomics Experiments." **CIKM 2025**. **Co-first author.**
4. Ziheng Duan, Tong Wu, **Xi Li**, Simon Dongbo Sun, Chaoyang Wang, Junhao Liu, Martin Renqiang Min, and Jing Zhang. "TargetSage: Identifying Therapeutic Target Genes with Interpretable and Robust LLM Reasoning." **NeurIPS 2026**, accepted poster.
5. **Xi Li**, Ziheng Duan, Yi Dai, Siwei Xu, and Jing Zhang. "iHAST: Integrating Hybrid Attention for Super-Resolution in Spatial Transcriptomics." **BMVC 2024**. *First author.*
6. Ziheng Duan, **Xi Li**, Zhuoyang Zhang, James Song, and Jing Zhang. "DISCO: A Diffusion Model for Spatial Transcriptomics Data Completion." **ICIP 2025**.

Manuscripts

7. Liyun Lin, Shuozhen Bao, **Xi Li**, Isaiah Yim, Dejiang Wang, Gaocai Li, Xiangjun Di, Shujie He, Junyi Xu, Kathryn Miller-Jensen, Jing Zhang, and Rong Fan. "Spatially Indexed Single-Cell Morphology and Transcriptome Profiling by DBiC-seq." **Nature Communications**, under review.
8. **Xi Li**, Ziheng Duan, Yi Dai, Yang Liu, and Jing Zhang. "Context Is Not a Side Note: A Support-Calibrated Reliability Framework for Biomedical Foundation Models." **KDD 2027**, under review. *First author.*
9. **Xi Li**, Ziheng Duan, Simon Dongbo Sun, Martin Renqiang Min, Yang Liu, and Jing Zhang. "Spine: Disentangling Tissue Context and Perturbation-Conditioned Response Fields in Spatial Genomics." **KDD 2027**, under review. *First author.*
10. "CCR2 Deficiency Attenuates Flow-Induced Carotid Artery Remodeling." **Vascular Pharmacology**, under review. Co-authors include Fu Gao and **Xi Li**; corresponding author: Yang Liu.
11. Fu Gao, Ian Mason, Mingze Dong, Yao Lu, Di Zhang, Xing Lou, **Xi Li**, Irbaz Hameed, Mingyu Yang, Mei Zhong, Markus Krane, Giovanni Ferrari, George Tellides, Yang Liu, Rong Fan, and Arnar Geirsson. "Cross-Species Multi-Omics Identifies a Conserved Fibrogenic Valvular Interstitial Cell Population across Mitral Valve Diseases." **Circulation**, submitted.